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PERFORMANCE SPECIFICATION

HORIZON REFERENCE SET, SHIP MOUNTED FOR LAMPS MK III HELICOPTER

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance and test requirements for a helicopter Horizon Reference Set (HRS).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, Code 414100B120-3, Highway 547, Lakehurst, NJ 08733-5100, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-S-901	-	Shock Tests HI (High Impact) Shipboard Machinery, Equipment and Systems, Requirements for
MIL-S-20708	-	Synchros, General Specification for
MIL-L-24560/1	-	Light, Aircraft Warning, Red, 120 Volts, 90 Watts, Six Lamps, Symbol 160.1

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II – Internally Excited)
MIL-STD-710	-	Synchros, 60 and 400 Hertz, Selection and Application of
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines
MIL-STD-1399/300-		Interface Standard for Shipboard Systems, Section 300A, Electric Power, Alternating Current (Metric)
MIL-STD-2036	-	General Requirements for Electronic Equipment Specifications

(Unless otherwise indicated, copies of the above specifications and standards are available from the Standardization Document Order Desk, 700 Robbins Avenue, Bldg 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government drawings form a part of this document to extent specified herein. Unless otherwise specified, the issue is that cited in the solicitation.

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DRAWINGS

NAVAIRWARCEN DRAWINGS

212602	-	Electronic Components Assembly (HRS)
212603	-	Control-Indicator (HRS)
212729	-	Horizon Reference Set Outline and Installation
520895	-	Stabilization Data Indicator Outline and Installation
520896	-	Electronic Components Outline and Installation
520897	-	Control-Indicator Outline and Installation
621154	-	Foundation Assembly (HRS)

(Copies of these drawings are available from the Commander, Naval Air Warfare Center Aircraft Division, Code 7252, Highway 547, Lakehurst, NJ 08733-5042.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1.1.

3.2 Systems description. The HRS shall provide a horizon reference which is independent of the ship's rolling motion. The visual display shall be a roll-stabilized, lighted reference bar to be located near the top of the hangar face of LAMPS MK III ships. The display shall be in the same plane of vision as viewed by a LAMPS MK III SH-60B helicopter pilot conducting a normal obstruction clearance scan during landing approach and recovery operation. The set shall be composed of three separate components (see figure 1).

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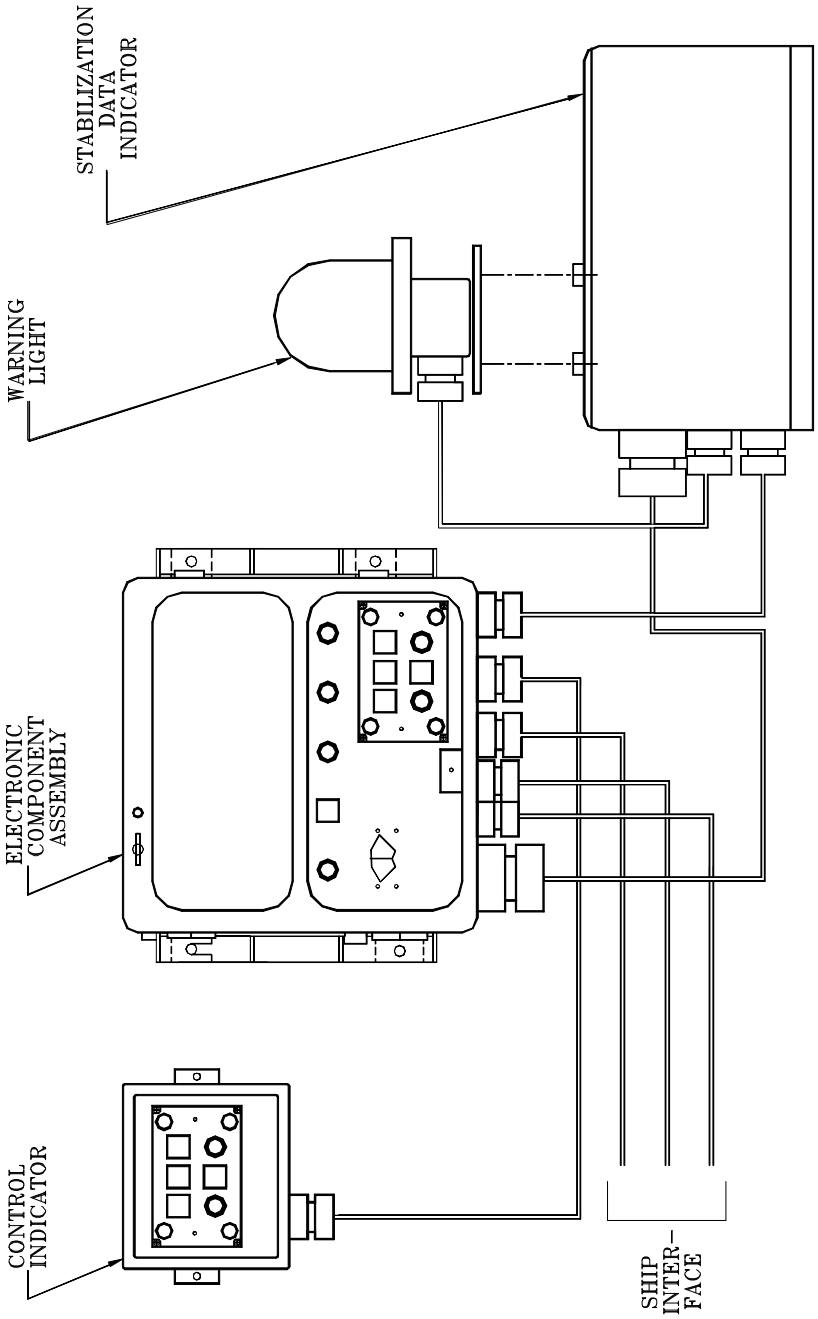


FIGURE 1. HRS components.

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a. A stabilization data indicator assembly composed of a lighted reference bar and a bar drive mechanism.

b. An electronic component assembly containing operating controls and indicators.

c. A control-indicator containing duplicate operating controls and indicators of the electronic component assembly for the remote control.

3.3 Performance characteristics. The HRS shall have the following performance characteristics.

3.3.1 Stabilization accuracy. The HRS shall be stabilized within the roll amplitude limits (see 3.3.3) to an accuracy within ±0.5° of the input signal.

3.3.2 Servo drive characteristics. The maximum angular velocity of the servo system driving the reference bar shall be not less than 20° per second, and the maximum angular acceleration under the same conditions shall be not less than 15° per second per second. The servo system damping ratio shall be not less than 0.70 or greater than 0.95. The servo system shall withstand wind velocities of up to 100 knots when subjected to the wind velocity test specified in 4.3.3.7.

3.3.3 Roll amplitude limits. The reference bar electro-mechanical drive shall incorporate adjustable roll amplitude limits, which shall provide for:

a. A minimum operating limit of not greater than ±17° and a maximum operating limit of not less than ±30° from the installation vertical. Adjustment increments between the minimum and maximum limits shall be not greater than 1°.

b. Inhibiting circuitry to limit reference bar rotation to not more than 1° greater than the set operating limit.

c. A positive reversal drive force if the reference bar rotation is more than 3° ± 1° greater than the set operating limit.

d. Mechanical stops to prevent further reference bar rotation greater than 8° ± 1° beyond the set operating limit.

3.3.4 System lock-up. The electro-mechanical drive system shall automatically align and lock the reference bar in a position parallel to the ship's deck when the HRS is shut down.

3.3.5 Reference bar lighting. The reference bar shall be illuminated using electroluminescent lamps conforming to 3.4.4. The length of electroluminescent lighting on the bar shall be not less than 9 feet 6 inches and the width shall be not less than two inches; lighting discontinuities

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between adjacent lamps shall be not greater than one inch. The intensity of the reference bar lighting shall be smoothly controlled from zero to maximum.

3.3.6 Fault detection and warning. The HRS shall incorporate provisions to monitor potential faults and provide automatic shut down and warning.

3.3.6.1 Fault detection. Monitoring shall detect the existence of any of the following faults:

a. Partial or complete loss of the input signal from the ship's gyro.

b. A differential, after initial stabilization, of greater than two angular degrees for more than two seconds between the input signal and the reference bar position when the reference bar is within operating limits (see 3.3.3).

c. Loss of 115 vac, 400 Hz input power to the set.

d. Partial or complete failure of the internal power supply.

3.3.6.2 Automatic shutdown. The presence of any of the system faults listed in 3.3.6.1 shall cause the HRS to automatically perform the following sequence:

a. Immediately turn OFF the reference bar illumination.

b. Immediately turn ON the red warning lamp.

c. For a two degree error (see 3.3.6.1b), automatically align and lock the reference bar in a position parallel to the installation platform and remove power from the servo system within ten seconds.

d. For all other fault conditions of 3.3.6.1, immediately remove power from the servo system and lock the reference bar in place.

3.3.6.3 Warning lamp. The warning lamp shall consist of an assembly conforming to MIL-L-24560/1 and shall have its primary location atop the housing of the stabilization data indicator assembly. The cable to the warning lamp assembly shall be not less than two feet long to allow for alternate installation of the warning lamp assembly on the stabilization data indicator assembly foundation (see 3.6.3). The intensity of the warning lamp shall be controllable from full brilliance to not less than 25 percent of the foot lambert output at full brilliance.

3.3.7 Operating control and indicators. The electronic components assembly and the control-indicator shall have the following switches, controls, and indicators:

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a. Power switch with local/remote indicators

b. Start switch with indicator

c. Stop switch with indicator

d. Reference bar stabilized indicator

e. Reference bar locked indicator

f. Fault indicator and reset switch

g. Reference bar illumination intensity control

h. Warning lamp intensity control

i. Panel and indicator illumination intensity control

In addition, the electronic component assembly shall have the following:

j. Display of reference bar position

k. Test controls and indicators

l. Indicators for 60 Hz and 400 Hz power for gyro signal reference voltage

3.3.7.1 Local/remote control. Design of the HRS shall provide for control of the installed set from either the electronic component assembly or the control-indicator if installed. If the control-indicator is not installed, the electronic component assembly shall provide sole control when optional interconnections are installed as specified on NAVAIRWARCEN Drawing 520896.

3.3.8. Physical characteristics. The HRS equipment shall conform to the interface criteria of figure 1. Weight and size of HRS components shall conform to the interface requirements of NAVAIRWARCEN Drawing 212729.

3.3.9 Reliability. The Mean Time Between Failures (MTBF) shall be equal to or greater than 500 hours (see 6.4.1). All failures shall be recorded.

3.3.10 Maintainability.

3.3.10.1 Mean Time To Repair (MTTR). For organizational maintenance

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(see 6.4.3), the MTTR shall be not greater than 15 minutes (see 6.4.2). Maximum maintenance time for correction of a failure shall be not greater than 45 minutes at the 95th percentile of distribution (see 4.3.7).

3.3.11 Environmental service conditions. HRS equipment shall operate as specified when subjected to the following environmental service conditions:

3.3.11.1 Temperature ranges. Exposed equipment (see 3.3.11.1.1) shall be subjected to a temperature range of -28 to +65 °C (-18 to +149 °F). Sheltered equipment (see 3.3.11.1.2) shall be subjected to a temperature range of 0 to +50 °C (32 to 122 °F). Exposed and sheltered, non-operating HRS equipment shall be subjected to a temperature range of -62 to +71 °C (-80 to +160 °F) (see 4.3.3.1 and 4.3.3.2).

3.3.11.1.1 Exposed equipment. The stabilization data indicator, including the illuminated reference bar and the warning lamp, shall be classified as exposed equipment.

3.3.11.1.2 Sheltered equipment. The electronic components assembly and the control-indicator shall be classified as sheltered equipment.

3.3.11.2 Humidity. HRS equipment shall be subjected to the humidity conditions in 4.3.3.3 and shall maintain specified performance parameters and show no evidence of physical deterioration.

3.3.11.3 Salt fog. The HRS components shall undergo the salt fog tests in 4.3.3.4. Electrical operation is required after the exposure period, and the HRS shall show no evidence of physical deterioration.

3.3.11.4 Sunshine. The electroluminescent lamps shall maintain specified performance and show no post test evidence of deformation, surface instability, splits, brittleness, or cracks in the lamp seal after the sunshine test of 4.3.3.5. Sunshine tests on complete assemblies are not required.

3.3.11.5 Fungus. Exposed and sheltered equipment shall operate and show no physical deterioration after testing in accordance with 4.3.3.6.

3.3.11.6 Wind. The HRS equipment shall not incur wind damage and shall operate as specified after being subjected to testing in accordance with 4.3.3.7.

3.3.11.7 Icing. HRS components shall operate as specified after being exposed to the icy conditions of 4.3.3.8. There shall be no physical damage to the equipment.

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3.3.11.8 Dust (fine sand). The HRS shall continue to operate as specified when exposed to the dusty environments of 4.3.3.9.

3.4 Electroluminescent lamps.

3.4.1 Electrical requirements. The lamps shall operate from a supply of 60 Hz to 400 Hz with variable voltage up to 130 vac.

3.4.2 Color. Each lamp shall emit green light with peak emissions at a wavelength of 540 nanometers ± 3 percent when operated at an ambient temperature of 23 ± 10 °C (73 ± 18 °F) from a 120 vac, 400 Hz supply. There shall be no color variance in the light with changes in supply voltage.

3.4.3 Luminance. Lamp luminance, when operated at an ambient temperature of 23 ± 10 °C (73 ± 18 °F) directly from a 120 vac, 400 Hz supply, and after a five hour luminance stabilization period, shall have an initial luminance of not less than 18 foot lamberts. After operating 1,000 hours, the lamp luminance shall be not less than eight foot lamberts. After operating 3,000 hours, the lamp luminance shall be not less than 4 foot lamberts. The luminance at any point of the luminous area shall be not greater than the luminance at any other point by more than 15 percent.

3.4.4 Lamp characteristics. Each lamp shall consist of the electroluminescent element housed in an enclosure capable of being bolted to a flat surface without stressing either the lamp or the mounting hardware. Lamp electrical leads shall be shrouded in sealant, projecting not more than 0.625 inch at the point of entry into the lamp. Lamps and electrical lead terminals shall be hermetically sealed and prevent incursion by air, gas, or moisture.

3.5 Electrical requirements. HRS equipment shall conform to the electrical requirements as follows:

3.5.1 Electrical power. The HRS shall operate on alternating current (ac) electrical power having characteristics as specified in MIL-STD-1399/300 for 400 Hz, type II or III and for 60 Hz, type I.

3.5.2 Vertical reference signal. The HRS shall accept the roll input signal from a type 18CX4, 400 Hz synchro control transmitter as specified in MIL-STD-710 and MIL-S-20708. The transmitter output is two electrical degrees for each degree of actual ship roll.

3.6. Mechanical design. HRS equipment shall conform to the mechanical design requirements as follows:

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3.6.1 Degree of enclosure. The degree of enclosure for exposed equipment (see 3.3.11.1.1) shall be watertight (see 6.4.5), and for sheltered equipment (see 3.3.11.1.2) shall be water controlled (see 6.4.4). Flow paths shall lead to a low point area equipped with drainage facilities to ensure complete overboard discharge by natural gravity flow.

3.6.2 External connections. The method and placement of external connections for HRS equipment shall be in accordance with NAVAIRWARCEN Drawings 520895, Sheets 2 and 3; 520896, Sheet 2; and 520897.

3.6.3 Equipment mounting. Equipment mounting provisions shall be in accordance with NAVAIRWARCEN Drawing 621154. The electronic components assembly and the control-indicator shall be designed for bulkhead mounting.

3.6.3.1 Shock and vibration. HRS components shall withstand shock and vibration environments. There shall be no decline in performance after testing (see 4.3.3.10 and 4.3.3.11).

3.6.4 Controls, indicators, and panel layouts. Controls and indicators for the electronic components assembly and for the control-indicator shall functionally conform to 3.3.7; panel illumination is required. In addition, the panel of the control-indicator shall be designed for non-illuminated areas. Panel layouts shall conform to NAVAIRWARCEN Drawing 212602 and 212603, Sheet 2. The electronic components assembly shall include a time totaling meter to indicate elapsed time for set operation and a key switch to disable the servo drive of the stabilization data indicator assembly for maintenance safety.

3.7 Thermal design. Thermal design shall encompass natural convection using inlet/outlet vents. Equipment design shall not require forced air or water cooling. HRS components shall not display any performance deterioration after testing in accordance with 4.3.4.1.

3.8 Identification and marking. Functional labeling on equipment shall be user friendly with respect to visibility and readability. Conductor and terminal end identification internal to equipment shall be conspicuously placed for electrical safety and information.

3.9 Workmanship. The HRS shall be clean in appearance. There shall be no burrs on threaded parts. Bearing assemblies shall have no surface defects, binding, or galling. Wiring shall be routed to avoid rough surfaces. Shielding/insulation shall prevent short circuits and leakage. Clearance between wires or cables and heat generating parts shall minimize deterioration.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.1.1).

b. Conformance inspection (see 4.1.2).

4.1.1 First article inspection. Unless otherwise specified, first article inspection shall include examinations and tests shown in table I. The first article shall be one or more representative system configurations composed of sample individual items selected from the first three production systems (see 6.2).

4.1.2 Conformance inspection. Unless otherwise specified (see 6.2), conformance inspection shall include the examinations and tests shown in table I. Conformance inspection shall be made on each item offered for delivery.

4.2 Performance of tests. The performance of tests shall be in accordance with MIL-STD–2036 and the following:

4.2.1 Failure criteria. Equipment or portions thereof, subjected to a test specified herein, shall be considered to have failed when any of the following occur:

a. Performance parameters exceed limits specified.

b. Catastrophic or structural failure.

c. Distortion or displacement of mechanical parts that cause difficulty in servicing or replacing a part.

d. Any condition that results in a hazard to personnel safety.

e. Deterioration, corrosion, or change in performance limits causing failure to meet operational service or maintenance requirements.

f. Deterioration of sealing compounds that would decrease service life or reliability.

g. Any condition that requires or results in a corrective maintenance action.

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TABLE I. Examination and tests.

Name or Subject of Test	Requirement Paragraph	Test Paragraph	First Article Inspection	Conformance Inspection
Group A: Non-Destructive Inspections				
Examination				
Weight & size	3.3.8	4.3.1	X	X
Components	3.2	4.3.1	X	X
Marking	3.8	4.3.1	X	X
Workmanship	3.9	4.3.1	X	X
Group B: Complex Non-Destructive Inspections & Tests				
Performance				
Full System	3.3	4.3.2.1	X	X
Individual Item	3.3	4.3.2.2	–	X
Environmental				
Temperature	3.3.11.1	4.3.3.1, 4.3.3.2	X	X
Enclosure	3.6.1	4.3.4	X	X
Thermal Design	3.7	4.3.4.1	X	–
Electrical	3.5	4.3.5	X	X
Reliability				
Demonstration	3.3.9	4.3.6.1	X	X
Maintainability				
Demonstration	3.3.10	4.3.7	X	–
Group C: Generally Destructive Tests				
Humidity	3.3.11.2	4.3.3.3	X	–
Salt Fog	3.3.11.3	4.3.3.4	X	–
Sunshine	3.3.11.4	4.3.3.5	X	–
Fungus	3.3.11.5	4.3.3.6	X	–
Wind	3.3.11.6	4.3.3.7	X	–
Icing	3.3.11.7	4.3.3.8	X	–
Dust (fine sand)	3.3.11.8	4.3.3.9	X	–
Shock	3.6.3.1	4.3.3.10	X	–
Vibration	3.6.3.1	4.3.3.11	X	X

4.3 Test methods. The following test methods of MIL-STD–2036 are applicable or modified as specified.

4.3.1 Surface examination. Equipment shall be inspected for visual defects, marking, workmanship, weight, and size.

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4.3.2 Performance tests.

4.3.2.1 Full system performance test. HRS equipment, installed in accordance with NAVAIRWARCEN Drawing 212729, Sheet 5 and combined into a system configuration, shall be tested to determine compliance with 3.3. This test should observe the conditions in 6.5.

4.3.2.2 Individual item performance tests. Individual equipment items not delivered as part of a complete equipment set shall be subjected to functional and performance tests at a number of points in the required operating range to detect deviations. Test points shall include the full range of operating controls or adjustments installed in the items. These tests should observe the conditions in 6.5.

4.3.3 Environmental tests.

4.3.3.1 Low temperature. The low temperature test shall be in accordance with MIL-STD-810, Method 502.3:

a. All HRS shall undergo Procedure I (storage) under severe cold, -51 °C (-60 °F).

b. Sheltered HRS shall undergo Procedure II (operation) under basic cold, -21 to –31 °C (-6 to –24 °F).

c. Exposed HRS shall undergo Procedure II under severe cold followed by Procedure III (manipulation - equipment set up and disassembly).

4.3.3.2 High temperature. The high temperature test shall be in accordance with MIL-STD-810, Method 501.3, Procedure II (operation) except as follows:

a. Step 4 - Applicable high temperature for the range specified shall be taken from the individual equipment specification.

b. The combined duration of steps 4 and 5 shall be not less than 12 hours, at least six of which shall be in the stabilized condition, prior to measuring performance parameters.

c. Step 7 - The equipment shall operate for a period of at least two hours in the stabilized condition before step 7 measurements are made.

4.3.3.3 Humidity. Humidity test shall be in accordance with MIL-STD-810, Method 507.3, Procedure I or II, as applicable.

a. Procedure I - natural humidity: HRS equipment open to the environment or frequently ventilated.

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b. Procedure II - induced humidity: Simulates unventilated conditions that may occur during storage or transit and simulates sealed items or items enclosed in sealed equipment.

4.3.3.4 Salt fog. The exposed equipment shall be tested in accordance with MIL-STD-810, Method 509.3, Procedure I (aggravated screening). If available test chambers preclude testing with the full size reference bar, testing may be conducted with a shorter bar representative of production fabrication with procuring activity approval.

4.3.3.5 Sunshine. The electroluminescent lamp on top of the reference bar shall be subjected to the sunshine test of MIL-STD-810, Method 505.3, Procedure I (cycling for heat effects-exposure in the open and in hot climates).

4.3.3.6 Fungus. The equipment shall be subjected to the fungus test specified in MIL-STD-810, Method 508.4.

4.3.3.7 Wind velocity. The stabilization data indicator assembly, with the reference bar and warning lamp installed, shall be mounted for operation with impinging wind forces. The equipment shall be operated with a sinusoidal input signal providing roll amplitudes of ±30° at a frequency of five cycles per minute while being subjected to a wind velocity of 75 knots directed normal to the reference bar in the horizontal plane. The equipment shall not incur damage during continuous operation under these conditions for at least one hour. If the stabilization data indicator assembly passes the 75 knot test, turn off the equipment, and increase the wind velocity to 100 knots directed normal to the reference bar in the horizontal plane for a period of at least 30 minutes. Upon completion of the 100 knot test, the equipment shall be visually inspected for damage, and turned on to check if the stabilization data indicator is operating as specified.

4.3.3.8 Icing. The stabilization data indicator assembly with the reference bar and warning lamp installed, in a non-operating condition, shall be subjected to a fresh water, ice forming spray until glaze ice build-up has reached 4.5 pounds per square foot of exposed surface. The ice build-up shall be maintained for a period of four hours. The temperature will then be increased to permit melting of the ice. Upon the completion of this test, the equipment shall be visually inspected for damage, and turned on to check if the stabilization data indicator is operating as specified.

4.3.3.9 Dust (fine sand). The stabilization data indicator assembly with the reference bar and warning lamp installed, shall be subjected to the dust tests as specified in MIL-STD-2036. No change in orientation of the equipment shall be required during the test. If the available test chambers preclude testing with a full size reference bar then testing may be conducted with a shorter bar representative of production fabrication providing prior approval is obtained from the procuring activity.

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4.3.3.10 Shock. HRS equipment shall be tested in accordance with MIL-S-901, grade A, class I, type A parameters. The stabilization data indicator assembly, with the reference bar and warning lamp installed, shall be mounted on a pedestal (see figure 2) such as NAVAIRWARCENLKE part number 621154-1 or equivalent.

4.3.3.11 Vibration. The equipment shall be subjected to MIL-STD-167-1, type I tests. The stabilization data indicator assembly, with the reference bar and warning lamp installed, shall be mounted on the same pedestal as used for the shock tests (see 4.3.3.10). The upper vibration frequency limit shall be 33 Hz.

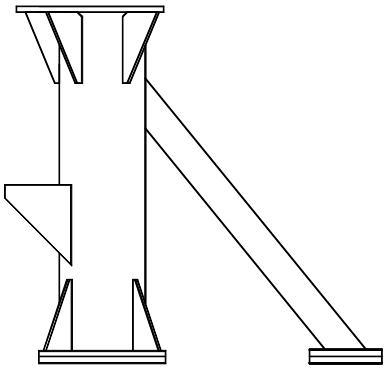


FIGURE 2. Shock test pedestal (stabilization indicator).

4.3.4 Mechanical and structural tests. These tests are limited only to the enclosure tests of MIL-STD–2036, and include watertightness and water control (see 3.6.1).

4.3.4.1 Thermal test. Operate the HRS at not less than 75 percent of rated value until thermal stabilization is achieved. Measure operating temperature of the parts that dissipate not less than 10 percent of the total power. Measure surface temperature of enclosures, front panels, and controls. Temperatures shall be within the ranges of 3.3.11.1. By visual examination determine if any of the HRS components have deteriorated due to thermal design.

4.3.5 Electrical tests. The electrical tests shall be completed in accordance with MIL-STD-2036, electrical design and construction section, and shall include steady state voltage, load characteristics, transient voltage and frequency, spike voltage, power interruption, insulation resistance, leakage current, equipment connections, electrostatic discharge (see 3.5).

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4.3.6 Reliability test. A reliability test shall be conducted as specified herein (see 3.3.9 and 6.6). The test shall be conducted subsequent to full system performance tests (see 4.3.2.1).

4.3.6.1 Test criteria. Testing of the reliability requirements (see 3.3.9) is based on the following test times and maximum failures:

Test time (hours)	Failures not to exceed
1945	1
2650	2
3350	3

Reliability tests shall be conducted under existing outdoor environmental conditions with the stabilization data indicator assembly exposed to weather. The other components of the system shall be sheltered so as to simulate shipboard installation with minimum shielding from outdoor temperature incursions. The test shall be conducted at nominal input voltages and frequencies and without vibration stressing. The equipment shall be in a "power OFF" condition for a minimum of two hours continuously during each calendar day of testing and shall be cycled ON and OFF the equivalent of once for every three hours of operation. During periods of equipment operation, control time shall be divided between the electronic components assembly and the control-indicator. Inputs to the system shall simulate expected ship roll motion amplitudes and frequencies.

4.3.6.2 Maintenance. Preventive maintenance during the reliability test shall be limited to that planned and approved for operational usage. Suspicion of incipient failure shall not be justification for replacing a part prior to its actual failure. If the failure of a suspected part may cause HRS failure, the part may be replaced with the approval of the procuring activity. A part so replaced shall be treated as a failed item and HRS failure shall be assumed to have occurred.

4.3.7 Maintainability demonstration. The contractor shall conduct a formal maintainability demonstration (see 3.3.10). The demonstration shall be performed on one of the first three production sets using maintenance procedures, support equipment and tools approved for use with the system (see 6.6).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (See 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain

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requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Horizon Reference Set (HRS) provides a stable, external visual, horizon reference to enhance the pilot's capability to operate LAMPS MK III helicopters in adverse weather conditions from the flight decks of LAMPS MK III ships experiencing extreme roll motions. The HRS is military unique equipment allowing safe interaction between ships and helicopters especially during extreme weather conditions unique to military operations.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1).

c. Whether first article inspection is required and disposition of tested equipment (see 3.1 and 4.1.1).

d. Any deviation from conformance inspection (see 4.1.2).

e. Items or sampling basis to be selected for conformance inspections as required (see 4.1.2).

f. Packaging requirements (see 5.1).

6.3 First article. When first article is required, the contracting officer should include specific instructions regarding examination arrangements, approval of first article test results and disposition of first articles. Invitations for bids should advise that the Government reserves the right to waive the requirement for first article inspection to those bidders offering a product, which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test data, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.4 Definitions. The following definitions apply in addition to those in MIL-STD-280.

6.4.1 Mean Time Between Failure (MTBF). MTBF is the total operating time in hours, divided by the total number of failures.

6.4.2 Mean Time To Repair (MTTR). MTTR is the total corrective maintenance time divided by the total number of corrective maintenance actions during a given period of time.

6.4.3 Organizational maintenance. Organizational maintenance includes all maintenance performed at, on, in, or to the HRS using only those skills, tools, support equipment, publications, procedures, and techniques specified and approved for normal service use when deployed.

6.4.4 Water control. Water control is the control of the flow paths of rain, wash water, or solvents, which enter non-watertight areas, to prevent standing water and eventual degradation of equipment performance and design service life.

6.4.5 Watertightness. Watertightness is the absolute prevention of water or water-solvent solution entry into a compartment or zone specified to be watertight when it is properly closed and secured for flight. Watertightness applies whether the aircraft is at rest, taxiing, or in-flight.

6.5 Test facilities, chambers, apparatus, and standard test conditions. Test facilities, chambers, apparatus, and standard test conditions used in conducting the tests specified herein should be in accordance with MIL-STD-2036. All equipment used to measure test results should conform to laboratory standards whose calibration is traceable to a calibrations source in accordance with ANSI-Z540.1.

6.6 Reliability and maintainability. For reliability test, consult MIL-HDBK-781 for guidance (see 4.3.6). For maintainability demonstration, consult MIL-HDBK-471, Appendix B.90, Test Method 8, “Test on a combined mean/percentile requirement” for guidance (see 4.3.7).

6.7 Corrective maintenance. Maintenance that occurs as a result of failures during first article testing may be used toward the maintainability demonstration requirement if performed in accordance with approved maintenance procedures and properly documented.

6.8 Subject term (key word) listing.

Helo Landing Aid, Visual
Landing Aid, Visual
Shipboard Alignment Aid, Visual, Helo
Visual Landing Aid, Helo

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6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due the extent of the changes.

CONCLUDING MATERIAL

Preparing Activity:
Navy - AS
(Project 1710-N062)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.
   NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-85281C(AS)	2. DOCUMENT DATE (*YYMMDD*) 20 July 1998

3. DOCUMENT TITLE
HORIZON REFERENCE SET, SHIP MOUNTED FOR LAMPS MK III HELICOPTER

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial: (2) DSN: (*If Applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY

a. NAME COMMANDER NAVAL AIR WARFARE CENTER AIRCRAFT DIVISION	b. TELEPHONE NUMBER (*Include Area Code*) (1) Commercial: (732) 323-2947 (2) DSN: 624-2947
c. ADDRESS (*Include Zip Code*) CODE 414100B120-3 HIGHWAY 547 LAKEHURST, NJ 08733-5100	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Logistics Agency (DLSC-LM), Attn: Carla Jenkins/John Tascher 8725 John J. Kingman Road, Ste 2533 Fort Belvoir, VA 22060-6221 Telephone (703) 767-6874

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